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204i

NOT NULL CONSTRAINT

Step 1

```
MariaDB [(none)]> use itc502;
Database changed
MariaDB [itc502]> create table Sales(
    -> Id int not null,
    -> Amount int not null,
    -> Vendor_Name varchar(255));
Query OK, 0 rows affected (0.330 sec)
```

Step 2

```
MariaDB [itc502]> insert into Sales values(1,22000, 'Amanda Harris');
Query OK, 1 row affected (0.163 sec)
```

```
MariaDB [itc502]> alter table sales
    -> modify Amount int null;
Query OK, 0 rows affected (0.520 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

MODIFY THE AMOUNT COLUMN INTO NULL

```
MariaDB [itc502]> insert into Sales values(2, null, 'Chris Jordan');
Query OK, 1 row affected (0.050 sec)
```

TABLE - Sales

```
MariaDB [itc502]> describe Sales;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| Id          | int(11)     | NO   |     | NULL    |       |
| Amount      | int(11)     | YES  |     | NULL    |       |
| Vendor_Name | varchar(75) | NO   |     | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
3 rows in set (0.073 sec)

MariaDB [itc502]> select * from Sales;
+----+--------+---------------+
| Id | Amount | Vendor_Name   |
+----+--------+---------------+
|  1 |  22000 | Amanda Harris |
|  2 |   NULL | Chris Jordan  |
+----+--------+---------------+
2 rows in set (0.001 sec)
```

UNIQUE CONSTRAINT

Step 1

```
Profit ...' at line 2
MariaDB [itc502]> create table Sales1(
    -> Id int not null unique,
    -> Amount int not null,
    -> Vendor_Name varchar(85),
    -> Profit Float);
Query OK, 0 rows affected (0.210 sec)
```

Step 2

```
MariaDB [itc502]> alter table Sales1
    -> add constraint UQ__Constrai Unique (Vendor_Name);
Query OK, 0 rows affected (0.225 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Step 3

```
MariaDB [itc502]> alter table Sales1 drop constraint UQ__Constrai;
Query OK, 0 rows affected (0.136 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Table Sales1

```
MariaDB [itc502]> describe Sales1
    -> ;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| Id          | int(11)     | NO   | PRI | NULL    |       |
| Amount      | int(11)     | NO   |     | NULL    |       |
| Vendor_Name | varchar(85) | YES  |     | NULL    |       |
| Profit      | float       | YES  |     | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
4 rows in set (0.033 sec)
```

CHECK CONSTRAINT

Step 1

```
MariaDB [itc502]> create table salary_info(
    -> ID int primary key,
    -> Name varchar(250) not null,
    -> Salary int check (Salary>10000));
Query OK, 0 rows affected (0.197 sec)
```

Step 2

```
MariaDB [itc502]> alter table salary_info
    -> add constraint CK_Salary
    -> Check (Salary > 10000);
Query OK, 0 rows affected (0.560 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Step 3

```
MariaDB [itc502]> alter table Salary_info
    -> drop constraint CK_Salary;
Query OK, 0 rows affected (0.028 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Table – salary_info;

```
ERROR 1146 (42S02): Table 'itc502.salary_inf0' doesn't exist
MariaDB [itc502]> describe salary_info;
+--------+--------------+------+-----+---------+-------+
| Field  | Type         | Null | Key | Default | Extra |
+--------+--------------+------+-----+---------+-------+
| ID     | int(11)      | NO   | PRI | NULL    |       |
| Name   | varchar(250) | NO   |     | NULL    |       |
| Salary | int(11)      | YES  |     | NULL    |       |
+--------+--------------+------+-----+---------+-------+
3 rows in set (0.012 sec)

MariaDB [itc502]> select*from salary_info;
Empty set (0.000 sec)
```

DEFAULT CONSTRAINT

Step 1

```
MariaDB [itc502]> CREATE TABLE salary_info1 (
    -> ID INT PRIMARY KEY,
    -> Name VARCHAR(250) NOT NULL,
    -> Salary INT,
    -> Salary_Date DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP);
Query OK, 0 rows affected (0.191 sec)
```

Step 2

```
MariaDB [itc502]> insert into salary_info1(ID, Name, Salary, Salary_Date
)Values(1, 'Andrew Buttler', 10350, '2021/07/22');
Query OK, 1 row affected (0.050 sec)
```

Step 3

```
MariaDB [itc502]> insert into salary_info1 (ID, Name, Salary) Values (2,
 'Nikola Harris', 9500);
Query OK, 1 row affected (0.244 sec)
```

Table – salary_info1

```
MariaDB [itc502]> describe salary_info1
    -> ;
+-------------+--------------+------+-----+---------------------+-------
+
| Field       | Type         | Null | Key | Default             | Extra
|
+-------------+--------------+------+-----+---------------------+-------
+
| ID          | int(11)      | NO   | PRI | NULL                |
|
| Name        | varchar(250) | NO   |     | NULL                |
|
| Salary      | int(11)      | YES  |     | NULL                |
|
| Salary_Date | datetime     | NO   |     | current_timestamp() |
|
+-------------+--------------+------+-----+---------------------+-------
+
4 rows in set (0.010 sec)

MariaDB [itc502]> select*from salary_info1;
+----+----------------+--------+---------------------+
| ID | Name           | Salary | Salary_Date         |
+----+----------------+--------+---------------------+
|  1 | Andrew Buttler |  10350 | 2021-07-22 00:00:00 |
|  2 | Nikola Harris  |   9500 | 2026-01-23 15:16:00 |
+----+----------------+--------+---------------------+
2 rows in set (0.001 sec)

MariaDB [itc502]>
```

SQL SERVER CONSTRAINTS

STEP 1

```
MariaDB [itc502]> create table cricketers(
    -> cricketer_id int primary key,
    -> last_name varchar(50) not null,
    ->  first_name varchar(50) not null,
    -> salary DECIMAL(10,2));
Query OK, 0 rows affected (0.183 sec)
```

STEP 2

```
MariaDB [itc502]> CREATE TABLE cricketers2 (
    -> cricketer_id INT,
    ->  last_name VARCHAR(50) NOT NULL,
    ->  first_name VARCHAR(50) NOT NULL,
    -> salary DECIMAL(10,2),
    -> CONSTRAINT cricketers2_pk PRIMARY KEY (cricketer_id));
Query OK, 0 rows affected, 1 warning (0.155 sec)
```

SQL SERVER FOREIGN KEY

```
MariaDB [itc502]> create table products (
    -> product_id int primary key,
    -> product_name varchar (50) not null,
    -> category varchar(25)
    -> );
Query OK, 0 rows affected (0.176 sec)
```

```
MariaDB [itc502]> create table inventory (
    -> inventory_id int primary key,
    -> product_id int not null,
    -> quantity int,
    -> min_level int,
    -> max_level int,
    -> constraint fk_inv_product_id foreign key(product_id) references products(product_id)
    -> );
Query OK, 0 rows affected (0.244 sec)
```

```
MariaDB [itc502]> create table products2
    -> (product_name varchar(50) not null,
    -> location varchar(50) not null,
    -> category varchar(25),
    -> Constraint products_pk primary key (product_name, location));
Query OK, 0 rows affected, 1 warning (0.197 sec)
```

```
MariaDB [itc502]> create table inventory2
    -> (inventory_id int primary key,
    -> product_name varchar(50) not null,
    -> location varchar(50) not null,
    -> quantity int,
    -> min_level int,
    -> max_level int,
    -> constraint fk_inv_product
    -> foreign key(product_name, location)
    -> references products2 (product_name,location));
Query OK, 0 rows affected (0.227 sec)
```

```
MariaDB [itc502]> describe product2
    -> ;
ERROR 1146 (42S02): Table 'itc502.product2' doesn't exist
MariaDB [itc502]>  describe products2
    -> ;
+--------------+-------------+------+-----+---------+-------+
| Field        | Type        | Null | Key | Default | Extra |
+--------------+-------------+------+-----+---------+-------+
| product_name | varchar(50) | NO   | PRI | NULL    |       |
| location     | varchar(50) | NO   | PRI | NULL    |       |
| category     | varchar(25) | YES  |     | NULL    |       |
+--------------+-------------+------+-----+---------+-------+
3 rows in set (0.013 sec)

MariaDB [itc502]> describe inventory2;
+--------------+-------------+------+-----+---------+-------+
| Field        | Type        | Null | Key | Default | Extra |
+--------------+-------------+------+-----+---------+-------+
| inventory_id | int(11)     | NO   | PRI | NULL    |       |
| product_name | varchar(50) | NO   | MUL | NULL    |       |
| location     | varchar(50) | NO   |     | NULL    |       |
| quantity     | int(11)     | YES  |     | NULL    |       |
| min_level    | int(11)     | YES  |     | NULL    |       |
| max_level    | int(11)     | YES  |     | NULL    |       |
+--------------+-------------+------+-----+---------+-------+
6 rows in set (0.013 sec)

MariaDB [itc502]> select*from products2
    -> ;
Empty set (0.001 sec)
```